

QUESTION DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to question according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-321 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

HYPOTHESIZE DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to hypothesize according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-322 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

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TEST DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to test according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-323 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

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REPLICATE DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to replicate according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-324 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

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MEASURE DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to measure according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-325 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

QUANTIFY DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to quantify according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-326 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

ANALYZE DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to analyze according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-327 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DIAGNOSE DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to diagnose according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-328 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MANAGE DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to manage according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-329 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

JSS 1-3 | Grid Experiment Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for treatments, controls and fair comparisons. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to recommend according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-330 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.